Computing and Software
SFWRENG 3A04
Software Design III - Large System Design
Winter 2025

McMaster University
ENGINEERING

Instructor Information

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Office Hours:
Wednesday: 2:00 PM to 3:00PM
I will be available at my office ITB-129

TA Information

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Class Times

Lectures: Three lectures per week
Labs: None
Tutorials:
The class will be split into three tutorial groups. Each tutorial group will have a two hour tutorial per week. Attendance at tutorials is compulsory. You MUST attend your assigned session of the tutorial. More information on the schedule and the organization of the tutorial will be posted on the course web-site.

Class Format

In Person

In-person attendance is required for this course.

Course Dates: 01/06/2025 - 04/08/2025
Units: 4.00
Course Delivery Mode: In Person
Course Description: Sustainable architectures; design for change and expansion; software architecture design space; object oriented analysis and design; architectural styles; methodology of making architecture decisions; project organization. Three lectures, one tutorial (two hours); second term Prerequisite(s): SFWRENG 3BB4; registration in any Software Engineering program

Instructor-Specific Course Information

The course provides a comprehensive coverage of software architectural styles. It provides students with a foundation that can be used to design large and sustainable software systems that exhibit high dependability and security qualities.

Upon completion of this course, participants should know and understand

1. Software requirements and its role in determining the architecture design
2. Software Architecture Design Space (Types of Software Structures, Software Elements, Software Connectors)
3. Design Principles and their role in leading to sustainable and dependable system

4. The main architectural styles (Batch Sequential, Pipe & Filter Architecture, Process-Control Architecture, Repository Architecture Style, Blackboard Architecture Style, Main/Subroutine Software Architecture, Master/Slaves Software Architecture, Layered Architecture, Virtual Machine, Non-Buffered
5. Event-Based
   Implicit Invocations, Buffered Message-Based Software Architecture, Model-View-Controller, Presentation-Abstraction-Control (PAC) Architecture, Client/Server, Multi-tier, Service-Oriented Architecture, Principles of Component-Based Design)
6. Heterogeneous Architecture (Methodology of Architecture Decision, Quality Attributes, Selection of Architecture Styles, Evaluation of Architecture Designs)

The course is quite practical in flavor and it does point to the use of the introduced concepts in software systems design. A certain degree of previous programming, design, and mathematical experience is essential for survival.

Meeting Details

The lectures and tutorials are all in-person. During the lecture time or during the tutorial, the instructor can give a surprise five to ten minute quiz. There will be **UP TO six** surprise quizzes.

During the office hour, the instructor will be available in ITB-129 (office).

Important Links

- Mosaic
- Avenue to Learn
- Student Accessibility Services - Accommodations
- McMaster University Library
- eReserves

It is the student's responsibility to be aware of the information on the course's Avenue to Learn page and to check regularly for announcements.

The latest version of the slides used in class, the material related to the course project, the material needed or used in the tutorial sessions, and announcements at the course website on Avenue to Learn.

Course Learning Outcomes

- Select among design methodologies
- Assess the impact of a requirement on the architecture design
- Identify any discrepancies in applying design principles in a proposed design
- Identify the applicable domain, benefits, and limitations of each one of the architectural styles
- Articulate the detailed design of the main components of each one of the architectural styles
- Enhance the sustainability of the designed system
- Trace a proposed architecture to the requirements
- Properly document a system from its requirement to its detailed design with emphasis on documenting the architecture
- Use Computer Aided Software Engineering (CASE) tools to carry through a term software project
- Organize and plan the development of a software system
- Evaluate design solutions regarding a specific set of requirements
- State and apply the design principles
- Select among system development documentation templates

Graduate Attributes

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

Course Schedule

The lectures will not necessary follow the order in which the topics are presented in the weekly breakdown given below. Regular class attendance is required.

A weekly breakdown of the course schedule

Topic	Chapters covered from textbook	Approximate Duration
1 Introduction to Software Architecture	Chapter 1	≈ 1 week
2 Software requirements and its role in determining the architecture design	Not covered by the textbook	≈ 1 week
3 Software Architecture Design Space (Types of Software Structures, Software Elements, Software Connectors)	Chapter 2	≈ 1 week
4 Models for Software Architecture (Architecture View Models, Architectural Description Languages (ADL))	Chapter 3	≈ 1 week

Topic	Chapters covered from textbook	Approximate Duration
5 Data Centered Software Architecture (Repository Architecture Style, Blackboard Architecture Style)	Chapter 6	≈ 1 week
Midterm Break		
6 Data Flow Architecture (Batch Sequential, Pipe & Filter Architecture, Process-Control Architecture)	Chapter 5	≈ 1 week
7 Hierarchy Architecture (Main/Subroutine Software Architecture, Master/Slaves Software Architecture, Layered Architecture, Virtual Machine)	Chapter 7	≈ 1 week
8 Interaction Oriented Software Architecture (Model-View-Controller, Presentation-Abstraction-Control (PAC) Architecture)	Chapter 9	≈ 1 week
9 Distributed Architecture (Client/Server, Multi-tier, Service-Oriented Architecture, Enterprise Service Bus)	Chapter 10	≈ 2 weeks
10 Heterogeneous Architecture (Methodology of Architecture Decision, Quality Attributes, Selection of Architecture Styles, Evaluation of Architecture Designs, Case Study: an Online Computer Vendor Component)	Chapter 12	≈ 1 week
11 Product Lines and Variants	Not covered by the textbook	≈ 1 week

Topic	Chapters covered from textbook	Approximate Duration
12 Architecture design for security	Not covered by the textbook	≈ 1 week

Required Materials and Texts

Textbook Listing: https://textbooks.mcmaster.ca

Software Architecture and Design Illuminated
ISBN: 9780763754204
Authors: Kai Qian, Xiang Fu, Lixin Tao, Chong-wei Xu, and Jorge Diaz-Herrera
Publisher: Jones and Bartlett Publishers
The textbook is available as an etext at McMaster Bookstore (see the following link):

https://campusstore.mcmaster.ca/CourseSearch/?course[]=UGRD,2251,SFWREN,SFWRENG+3A04,C01&

Price: Buy Digital (CampusEbookstore) $81.95

Course Evaluation

Project

A major component of this course is a team project. The milestones of the project are marked by four deliverables that deal with (1) the requirements, (2) the architecture design, (3) the detailed design, and (4) construction of a first team-prototype, and the final documentation and the construction of the final team system.

Each team agrees on a software architecture and interfaces. The members of the team complete a system adhering to those interfaces and must demonstrate a working system. A team must submit a unique deliverable per due date. Within a team, the grade assigned to a member is proportional to the member's contribution. With each deliverable, you must provide a statement that details the contribution of each team member and that is signed by all team members. The instructor reserves the right to re-evaluate the contribution of a student (to a single deliverable or to a set of deliverables) via an oral examination. After a re-evaluation of the contributions of a team member, their previously assigned grades can be changed.

Students are encouraged to use freely available tools for revision and version control, and for project organization and management.

The following table gives a tentative timetable for the deliverables of the project.

Deliverable	Tentative Due Date	Deliverable Weight
D1	February 14	10%
D2	March 07	10%
D3	March 21	10%
D4	Week of April 07 (A timetable for demos will be set by the TAs)	10%

Your deliverables are due online by 11:59 PM on the due date. There will be 48 hours of grace and deliverables submitted after the grace period will be marked with a late penalty of 20% of the full grade per day.

When evaluating a team's report for a deliverable, a TA might deem that the report is unsatisfactory and cannot be used for the next stage of system design. In this case, you are encouraged to work with the TA on improving your deliverable.

Any request for remarking must be first directed to the TA that has marked your work. After you have talked to your TA and still believe that you deserve a higher mark, then you can contact the instructor. When the instructor re-assesses a work or an exam, all the work aspects or exam questions will be reconsidered.

Surprise quizzes

During the lecture time or during the tutorial, the instructor can give a surprise five to ten minute quiz. There will be **UP TO six surprise quizzes**. If the class writes one quiz during all the term, it counts for 2% of the final grade. If the class writes more than one quiz, the one with the lowest grade does not count, while each one of the others count for 2% of your final score. If, for reasons accepted by the Associate Dean's Office, you write only one quiz, only that quiz counts. If for accepted reasons you miss all the quizzes, your final exam will be worth 40%. **There are no deferred quizzes.**

Midterm exam

There will be one midterm exam. It will take place on **Monday, March 3, from 11:30 AM to 12:20 PM**. It will be 50 minutes in duration and will cover material from the lectures, tutorials, project, and from the required textbook. The midterm exam is worth 20% of the final grade.

Final exam

The final examination will be scheduled by the Registrar's office in the usual way. It will be three hours in duration and will cover the material of the lectures, tutorials, project, and of the required textbook. The final exam counts for *(40 - 2 × Number_of_Quizzes_that_Count)%* of your final score. For example, if the class writes 4 quizzes, only 3 quizzes count and the final exam will be worth (40-2*3)%.

IMPORTANT NOTE:

To pass the course, you must gain a grade of at least 50% on the midterm and the final exam, or on the final exam only. In other terms, you must have the TWO following conditions satisfied (Satisfying only one of the two below conditions is not sufficient to pass the course):

1. Max**(** (((Final_Exam_Grade + Quizzes_Grades)/40)*100)**,** ((((Final_Exam_Grade + Quizzes_Grades)/ + Midterm_Exam_Grade)/60)*100) **)** >= 50
2. Final_Exam_Grade + Quizzes_Grades + Midterm_Exam_Grade + The grades of the project deliverables > = 50

The midterm exam will be marked over 20 points and the final exam will be marked over 40 points.

If you pass the course (i.e., your grades satisfy the above conditions 1 AND 2), the course final grade is calculated as follows:

Course_Final_Grade = Final_Exam_Grade + Quizzes_Grades + Midterm_Exam_Grade + The grades of the project deliverables.

The instructor reserves the right to conduct any deferred midterm or final exams orally.

Course Evaluation Details

Group Work: While different group members may be responsible for different parts, that doesn't mean you should do it entirely independently and not bother to look at each other's work. It shouldn't be obvious to the reader that different parts were written by different people. You should be reading each other's work and giving feedback to each other.

Grading Scale

The McMaster 12 Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Late Assignments

It is essential that you meet the deadlines for the deliverables; there is no credit for material submitted after the deadline.

Absences, Missed Work, Illness

1. Missed deliverable or exam: A zero will be entered for the missed work unless the instructor receives a notification from the Associate Dean office or an MSAF within a reasonable time. It is YOUR responsibility to follow up with your instructor immediately (normally within 2 working days) to discuss possible consideration. **The course project components cannot be MSAFed.**
2. Tutorials: The class will be split into three tutorial groups. Each tutorial group will have a two hour tutorial per week. Attendance at tutorials is compulsory. You MUST attend your assigned session of the tutorial. More information on the schedule and the organization of the tutorial will be posted on the course web-site.
3. The instructor reserves the right to adjust the grades for a deliverable, or the final exam by decreasing every score by a fixed number of points.
4. The instructor reserves the right to increase by a fixed number of points the final scores of all the students who handed in all the deliverables and attended all the tutorials of the term (except if the student misses a tutorial for a justifiable reason).
5. The instructor reserves the right to assign extra grades for extra work done by willing students. However, the work subject to extra grades should be advertised during the lectures.
6. A remarking request of a deliverable is considered by the TAs and the instructor only if it is made within the two weeks that follow the return date of the majority of the concerned deliverable.
7. The instructor reserves the right to require a deferred exam to be oral.
8. No responsibility for loss of deliverables can be assumed by either the instructor or the Teaching Assistants. Keep copies of your own contributions to the project, your reports, and your email exchanges with your team members.
9. Calculators are not needed for this course and their use will not be permitted during exams.

10. The lectures will not necessary follow the order in which the topics are presented in the detailed course outline. Regular class attendance is required.
11. Significant study, reading, and group discussions outside of class and tutorials are required.
12. Suggestions on how to improve the course and the instructor's teaching methods are always welcomed.

Generative AI: Some Use Permitted

Students may use generative AI for ONLY editing, translating, outlining, brainstorming, or revising their work throughout the course so long as the use of generative AI is referenced and cited following citation instructions given in the syllabus. Use of generative AI outside the stated use of editing, translating, outlining, brainstorming, or revising without citation will constitute academic dishonesty. It is the student’s responsibility to be clear on the limitations for use and to be clear on the expectations for citation and reference and to do so appropriately.

Here are some guidelines for citing generative AI:

- **Author:** Use the creator of the AI as the author. For example, "OpenAI" or "Google".
- **Date:** Include the date the content was generated.
- **Title:** Use the name of the AI tool, such as "ChatGPT" or "Gemini".
- **Version:** Include the version of the AI tool, if available.
- **Description:** In brackets, clarify the type of generative AI, such as "large language model".
- **Location:** Include the URL for the tool, or the URL for the specific content if possible.
- I**n-text citations:** Include the name of the AI tool, its owner, and the year of publication.
- **Reference list:** Include further details of how you used the tool, such as the prompt you provided and what the generated text offered.

- **Chicago style:** Cite the AI conversation like a personal communication, under the name of the publisher or developer. Include a publicly available URL.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Equity, Diversity, and Inclusion

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or to contact the Equity and Inclusion Office.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

Academic Advising

For any academic inquires please reach out to the Office of the Associate Dean (Academic) in Engineering located in JHE-Hatch 301.

Details on academic supports and contact information are available from:

https://www.eng.mcmaster.ca/programs/academic-advising

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the Policy on Requests for Relief for Missed Academic Term Work.

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.